What Is the Influence of Task Order in Continual Learning? An Empirical Study and Order-Aware Evaluation for Motion Forecasting

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***Abstract*—Aggregate scores for continual learning algorithms presuppose a distribution over task orders that is seldom made explicit. We formulate algorithm comparison as an expectation over a declared order distribution, which separates training variation from order variation, and apply it to motion forecasting on four road-layout tasks from the INTERACTION dataset, with all 24 task orders, four algorithms, and three training replicates (288 runs). Task order accounts for 65% to 91% of the variation in current-task error and backward transfer, but 90% to 99% of that share is the identity of the starting task, because the smallest task cannot be learned from a cold start. Beyond the first stage, the order effect is comparable to training variability, and most reversals of algorithm rankings between orders lie within it. Backward transfer depends on the order largely through its reference term, which weights the first task most heavily, so relative improvement and final error rank the algorithms differently. A transition-direction descriptor used to weight orders mainly indexes the position of the smallest task and leaves the ranking unchanged. We recommend that benchmarks state the order distribution behind any aggregate, report order-resolved and starting-task-resolved outcomes with warm-start and final error, and fix any weighting rule before outcomes are examined.**

***Index Terms*—Benchmark testing, continual learning, evaluation protocol, motion forecasting, task order.**

I. Introduction

CONTINUAL learning (CL) acquires knowledge from a sequence of tasks under changing data distributions, and updates for a new task can interfere with representations required by earlier tasks [1]–[3]. Replay, regularisation, and gradient constraints counter this interference through different retention mechanisms [4]–[8], so their outcomes depend on the learning history fixed by the task order as well as on the task set.

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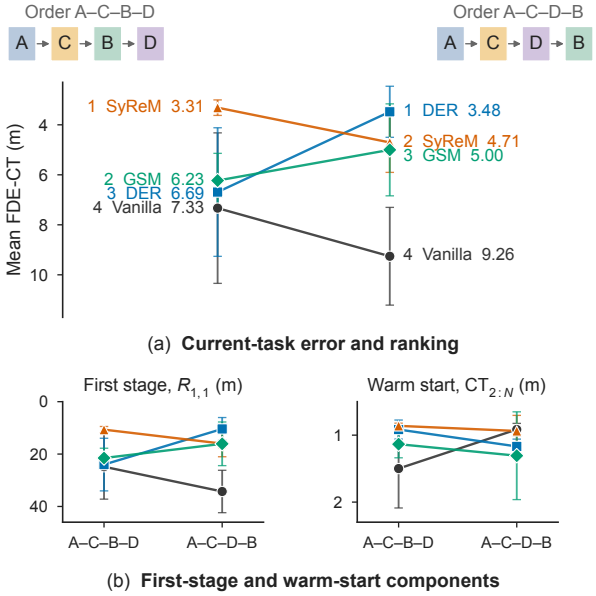


Fig. 1. Two task orders that differ only in their final two tasks rank the four algorithms differently. (a) Replicate means of the current-task final displacement error (FDE-CT) with bars spanning one sample standard deviation; in both orders the leader's gap to the runner-up exceeds the pooled replicate standard deviation. (b) The same runs split into the first-stage error $R_{1,1}$ and the warm-start error $\mathrm{CT}_{2:N}$: the reversal arises in the first stage.

Any aggregate score for a CL algorithm therefore presupposes a distribution over the orders in which the tasks were presented. Many evaluation protocols leave that distribution implicit: they repeat one or a few prescribed orders while varying initialisation or within-task sample order [8], [9], which places all comparison weight on those orders. Studies of order-dependent forgetting, order-wise performance disparity, ordering principles, and sequence transferability show that task transitions influence retention and transfer [10]–[13]. These studies explain or select orders; they do not report how the variation in outcomes over a complete set of $N!$ permutations divides between order, algorithm, their interaction, and training randomness, or which property of an order carries the order effect.

This paper asks how much of the variation in CL outcomes on a fixed task set is attributable to task order, which property of an order carries that effect, and whether algorithm rankings persist across orders. It proposes no new algorithm. Algorithm

comparison is formulated as an expectation over an explicit order policy, with the uniform policy as the equal-order reference. The sum of squares of each metric is decomposed into order, algorithm, interaction, and within-cell training components, the order term is split by starting task, and a replicate bootstrap attaches a probability to every leading algorithm. A warm-start current-task error and a decomposition of backward transfer into subsequent error and initial reference isolate the first learning stage. A transition-direction descriptor computed from task statistics defines a descriptor-weighted policy as a worked example of a nonuniform policy applied identically to all algorithms (Fig. 2).

Continual motion forecasting across traffic environments [14], [15], for which evaluation under varying data-stream sequences has been recommended [16], provides the empirical setting, with the UQnet endpoint-heatmap backbone [17]. Four tasks defined by road layout in the INTERACTION dataset are learned in all $4! = 24$ orders by Vanilla sequential fine-tuning and three memory-based algorithms, dark experience replay (DER), synergetic memory rehearsal (SyReM), and gradient scenario memory (GSM), with three training replicates per order and algorithm (288 runs). Fig. 1 previews the central finding: two orders that differ only in their last two tasks change the leading algorithm by more than the replicate variability, but the reversal arises in the first learning stage, and beyond it the algorithms are not separable.

The main contributions are summarised as follows.

1) *Formulation and benchmark.* The expectation over an explicit order policy defines a common comparison target, and the law of total variance separates within-order training variation from across-order variation. All 24 orders are enumerated with released per-run results, and the decomposition, adjustment, and bootstrap procedures form a reusable analysis toolkit.
2) *Empirical finding.* Task order accounts for 65% to 91% of the variation in current-task error and backward transfer, but 90% to 99% of that order effect is the identity of the starting task, which acts through the cold start of the smallest task. Once the first stage is removed, the residual order effect is comparable to training variability; the per-order leader is separable from the runner-up in only 2 to 11 of the 24 orders for current-task error and backward transfer, against 16 or 17 of 24 for final error.
3) *Metric diagnosis.* The first task enters the reference term of stage-averaged backward transfer with weight $11/18$ for four tasks, so a poor first fit raises current-task error and drives backward transfer towards large negative values at once; relative improvement and absolute final error consequently rank the algorithms differently. The transition-direction descriptor mainly indexes the position of the smallest task in this task set, and the descriptor-weighted policy leaves the full ranking unchanged.

II. Related Work

A. Task Order and Curriculum Effects

Curriculum learning investigates the effect of example or task presentation on optimisation [18]. In CL, ordering also affects retention of previous knowledge. Bell and Lawrence [10] examine order-dependent forgetting and task-distance heuristics; Li and Hiratani [12] derive ordering principles in a teacher–student model. Analyses of task similarity and overparameterisation further characterise interference [19]. Nguyen *et al.* [20] relate error rates to the total complexity of a task sequence and to its sequential heterogeneity, the sum of distances between consecutive tasks; the path length used in Section IV-D is of this form.

Yoon *et al.* [11] quantify order-wise performance disparity, establishing order fairness as an evaluation criterion. Measures of sequence transferability relate task transitions to performance using learned predictors [13]. The present analysis uses marginal task statistics to define a shared order policy and asks how much of an observed ranking survives a change of order.

B. Continual Learning for Motion Forecasting

Continual forecasting addresses adaptation across changing traffic environments [16] through generative replay [21]–[23], regularisation under distribution shift [24], and task-free streams without supplied task boundaries [9], [25], [26].

D-GSM uses traffic divergence and scenario-organised memory to constrain gradient updates [14], H2C combines diversity-based selection with equiprobable replay [9], SyReM integrates a memory-loss constraint with gradient-based rehearsal selection [15], and Dual-LS coordinates long- and short-term knowledge through complementary rehearsal [27]. These studies evaluate each method along one or a few task orders; the dependence of the resulting ranking on the order has not been examined.

C. Evaluation Protocols for Continual Learning

Average performance over all tasks, current-task performance, and backward transfer are the standard quantities of continual learning evaluation [6], [28], and the appropriate protocol differs between task-, domain-, and class-incremental settings [29]. Early protocol studies showed that reported forgetting depends on the evaluation design, including the task sequences used [30], [31], and proposed metric sets beyond final accuracy [32]. Hyperparameter selection on the evaluation scenario can bias comparisons [33], and unequal update costs can alter conclusions about online CL [34]. Recent benchmarks treat the order either as an adversarial choice to be searched [35] or as a sampled factor whose uncertainty is reported [36].

Order coverage is therefore a protocol choice. We distinguish repeated training within an order from evaluation across orders, separate the starting task from the remaining order, and assess backward transfer alongside initial, warm-start, and final error.

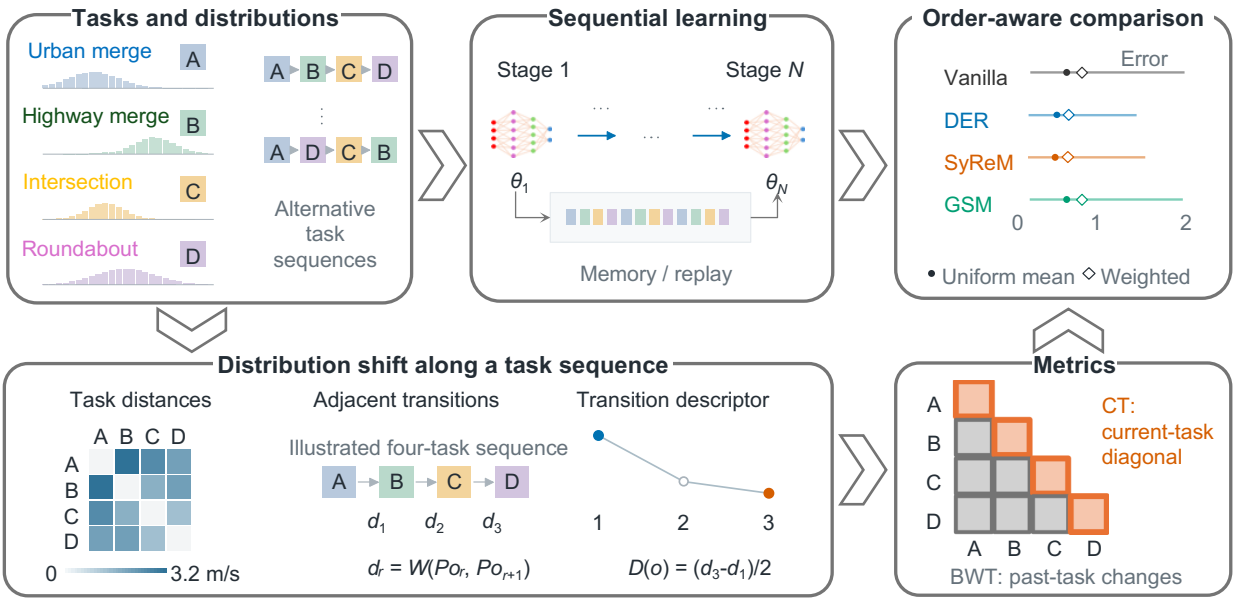


Fig. 2. Order-aware evaluation workflow. Task distributions define transition descriptors; sequential training provides current-task and backward-transfer outcomes. A shared order policy aggregates these outcomes for algorithm comparison. The distributions, networks, and comparison marks are schematic; d_r denotes the W1 speed distance between consecutive tasks and θ_r the parameters within the state z_r.

III. Problem Formulation

A. Sequential Learning Conditioned on Tasks

Let $[N] = \{1, \dots, N\}$ index a task set $\mathcal{T}$. Task j is specified by a distribution $\mathcal{D}_j$ on $\mathcal{X} \times \mathcal{Y}$ and a finite training set $\mathcal{S}_j = \{(\mathbf{X}_{jk}, \mathbf{Y}_{jk})\}_{k=1}^{n_j}$. The tasks share the input and output spaces, while their data distributions may differ. An order (task sequence) $o = (o_1, \dots, o_N)$ belongs to the permutation group $\mathfrak{S}_N$, with $|\mathfrak{S}_N| = N!$. Evaluation uses a common subset $\mathcal{O} \subseteq \mathfrak{S}_N$ for every algorithm.

For algorithm $a \in \mathcal{A}$, let $z_r^{(o,a,s)}$ denote its state after stage r in training replicate s, comprising model parameters, optimiser state, and algorithm-specific persistent information such as the replay memory $\mathcal{M}_r^{(o,a,s)} \subseteq z_r^{(o,a,s)}$. Sequential learning is represented by

$$z_r^{(o,a,s)} = \mathcal{U}_a\big(z_{r-1}^{(o,a,s)}, \mathcal{S}_{o_r}; \xi_r^{(o,a,s)}\big), \tag{1}$$

where $\mathcal{U}_a$ is the update operator and $\xi_r^{(o,a,s)}$ denotes the stochastic inputs for that run and stage; the initial state is likewise run-specific. The memory satisfies $|\mathcal{M}_r^{(o,a,s)}| \le B_a$ for a specified record capacity B_a, a record may include a stored model output, and historical training examples are accessible only through this memory. Task boundaries are provided, all tasks use a shared prediction head, and no single-pass training requirement is imposed.

B. Risk and Knowledge Retention

For loss ℓ and predictor $f_{\boldsymbol{\theta}}$, the risk on task j is

$$\mathcal{R}_j(\boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{X},\mathbf{Y})\sim\mathcal{D}_j}[\ell(f_{\boldsymbol{\theta}}(\mathbf{X}), \mathbf{Y})]. \tag{2}$$

A reference objective for stage $r > 1$ is

$$\boldsymbol{\theta}_r^\star \in \arg\min_{\boldsymbol{\theta}} \left\{ \mathcal{R}_{o_r}(\boldsymbol{\theta}) + \frac{\lambda_r}{r-1} \sum_{i<r} \mathcal{R}_{o_i}(\boldsymbol{\theta}) \right\}, \tag{3}$$

where $\lambda_r \ge 0$ balances current-task fit and past-task risk. Stage 1 contains only the current-task term, so the first stage is single-task learning from the initial state. Replay, regularisation, and gradient constraints approximate the historical risk through different mechanisms [4]–[6], [8]. Absolute historical risk and its change since initial learning are distinct quantities, hence separate measures of final error and backward transfer.

C. Evaluation over Task Orders

Let $Z_m(o, a, \xi)$ be the scalar outcome of a complete run under metric m, where ξ collects initialisation and subsequent training randomness for the specified order and algorithm. For an order policy q satisfying $q(o) \ge 0$ and $\sum_{o\in\mathcal{O}} q(o) = 1$, the comparison target is

$$\mu_m(a; q) = \sum_{o\in\mathcal{O}} q(o)\, \mathbb{E}_\xi[Z_m(o, a, \xi)]. \tag{4}$$

For S training replicates, define $y_m(o, a) = S^{-1} \sum_{s=1}^{S} Z_m(o, a, \xi^{(o,a,s)})$. The empirical policy score is

$$\widehat{\mu}_m(a; q) = \sum_{o\in\mathcal{O}} q(o) y_m(o, a). \tag{5}$$

All algorithms share $\mathcal{O}$, q, and the metric definition. For fixed a, m, and q, assume Z_m has a finite second moment and draw $o \sim q$. The law of total variance separates the two sources of variability:

$$\begin{aligned}\mathrm{Var}(Z_m) = {} & \mathbb{E}_{o\sim q}[\mathrm{Var}_\xi(Z_m \mid o)] \\ & + \mathrm{Var}_{o\sim q}(\mathbb{E}_\xi[Z_m \mid o]).\end{aligned} \tag{6}$$

The first term describes stochastic training variation within orders; the second describes variation in expected performance across orders. Repetition of a single order estimates only the

first component, so order-aware evaluation requires an explicit order policy together with order-resolved outcomes.

For algorithms a and b, let $\delta_m(o; a, b) = y_m(o, a) - y_m(o, b)$ be the measured difference. Equation (5) gives

$$\widehat{\mu}_m(a; q) - \widehat{\mu}_m(b; q) = \sum_{o \in \mathcal{O}} q(o)\delta_m(o; a, b). \tag{7}$$

A common sign of δ_m across orders preserves the weak ranking under every policy supported on $\mathcal{O}$, whereas opposite signs permit a reversal under policies concentrating on different orders; order-wise differences thus distinguish uniform dominance from dependence on the evaluation distribution. This property concerns measured means, and replicate variability is addressed in Section IV-D.

IV. Order-Aware Evaluation

A. Distances Between Task Distributions

We describe each task through empirical distributions of target-agent speed and acceleration. Speed is the norm of the two velocity components at each of nine historical time steps. Acceleration is the adjacent speed difference divided by 0.1 s, retaining values in $[-5, 5]$ m/s^2. States are pooled within each task to form the empirical distributions, which therefore contain correlated time-step observations.

The candidate measures probe complementary aspects of distribution shift. Wasserstein-1 (W1) quantifies displacement of probability mass along the physical feature axis [37]. Kullback–Leibler (KL) divergence measures directed probability mismatch, whereas Jensen–Shannon (JS) divergence provides a symmetric, bounded comparison of the same histograms [38]. Maximum mean discrepancy (MMD) compares kernel mean embeddings without histogram discretisation [39]. Applying all four measures to speed holds the feature representation fixed while varying the notion of discrepancy; applying W1 additionally to acceleration examines a different aspect of motion with the same distance functional.

For feature $f \in \{\text{speed}, \text{acceleration}\}$, let F_i^f denote its empirical cumulative distribution function in task i. We compute

$$d_{ij}^{\mathrm{W}}(f) = \int_{\mathbb{R}} |F_i^f(t) - F_j^f(t)| \, \mathrm{d}t. \tag{8}$$

This calculation uses every retained observation, without subsampling or standardisation. Its units are m/s for speed and m/s^2 for acceleration.

KL and JS divergences are computed on $H = 100$ shared equal-width speed histograms with additive smoothing, and MMD on a fixed standardised sample of 2,000 speed observations per task with a Gaussian kernel; the estimators are specified in Appendix A-A. KL retains its direction; W1, JS, and MMD matrices are symmetric with zero diagonals, and scaling to $[0, 1]$ is applied only to plotted matrices.

The primary distance is W1 on speed, which has an interpretable physical scale. This choice emerged from an exploratory scan of 20 distance representations and 15 aggregation operators against the CL outcomes (Appendix A-B). We report the five candidate combinations under fully specified estimation settings and retain the original choice without further outcome-based retuning.

B. A Descriptor of Transition Direction

For order o, let $\Delta_r(o) = d_{o_r, o_{r+1}}$ be the distance at the rth transition. The descriptor is its least-squares slope against transition position:

$$D(o) = \frac{\sum_{r=1}^{N-1}(r - \bar{r})\Delta_r(o)}{\sum_{r=1}^{N-1}(r - \bar{r})^2}, \qquad \bar{r} = N/2. \tag{9}$$

It is defined for $N \geq 3$. For the four-task experiment this reduces exactly to

$$D(o) = \frac{d_{o_3, o_4} - d_{o_1, o_2}}{2}. \tag{10}$$

The middle transition has zero coefficient. A positive value means that the final transition is larger than the first; it does not imply that all transitions increase, or that individual tasks become intrinsically harder. We therefore refer to D as a transition-direction descriptor, and its interpretation as objective-specific difficulty is an empirical association to be checked.

For symmetric distances, $D(o^{\mathrm{rev}}) = -D(o)$, where o^{rev} reverses the order. There are only three partitions of four tasks into two unordered pairs. Each partition gives two signed contrasts, so (10) has at most six distinct values across the complete order set. Swapping the tasks within either endpoint pair leaves D unchanged even though it can change training. These ties are retained in the correlations and weighting, and order-resolved results reveal the variation among orders that share a descriptor value.

The slope is equivariant to a common positive affine transformation of the distances, $d'_{ij} = ad_{ij} + b$ with $a > 0$ giving $D'(o) = aD(o)$, so its rank and the min–max weights below are unchanged by a common linear rescaling; an arbitrary monotone transformation does not preserve them, because the slope is a difference of distances.

A further property of this task set matters for interpretation. The first and last transitions of any order form one of the three pair partitions of the task set, and in every partition the pair containing task A has the larger speed distance ($d_{AB} = 3.25 > d_{CD} = 1.36$, $d_{AC} = 2.63 > d_{BD} = 2.15$, and $d_{AD} = 2.12 > d_{BC} = 1.74$ m/s; the five candidate matrices are released with the data), so the sign of D in (10) is determined by where task A appears: $D(o) < 0$ exactly when A occupies position 1 or 2, for all 24 orders, and the Spearman correlation between D and the position of A is 0.79. Associations with D are therefore examined alongside associations with the position of A in Section VI-C.

C. Uniform and Descriptor-Weighted Policies

The uniform policy $q(o) = 1/|\mathcal{O}|$ assigns equal probability to each evaluated order and gives the equal-order reference

$$\bar{y}_m(a) = \frac{1}{|\mathcal{O}|} \sum_{o \in \mathcal{O}} y_m(o, a). \tag{11}$$

It is reported alongside the minimum and maximum order means, per-order ranks, and training variation. A deployment-specific comparison requires a correspondingly specified order distribution.

For a supplementary descriptor-weighted policy, let $D_{\min} = \min_{o\in\mathcal{O}} D(o)$ and $D_{\max} = \max_{o\in\mathcal{O}} D(o)$, and define

$$\widetilde{D}(o) = \frac{D(o) - D_{\min}}{D_{\max} - D_{\min}}. \tag{12}$$

When $D_{\max} = D_{\min}$, uniform weights are used and the weighted result equals (11). Otherwise, the directions of association observed in the exploratory analysis (Section VI-C) fix the common weights for the current-task error (CT) and backward transfer (BWT), both defined in Section V-C,

$$w_{\mathrm{CT}}(o) = 1 - \widetilde{D}(o), \qquad w_{\mathrm{BWT}}(o) = \widetilde{D}(o), \tag{13}$$

and, for each metric separately,

$$\bar{y}_m^w(a) = \frac{\sum_o w_m(o) y_m(o,a)}{\sum_o w_m(o)}. \tag{14}$$

The normalised weights define $q_m(o) = w_m(o)/\sum_{o'} w_m(o')$ in (5); the same policy applies to every algorithm, and metrics with incompatible units are never combined. Because the complete order set contains the reversal of every order, a nonconstant symmetric-distance descriptor gives $D_{\min} = -D_{\max}$, so both weight vectors sum to $|\mathcal{O}|/2$. The policy emphasises descriptor regions associated with larger errors or less favourable backward transfer but provides no worst-case guarantee, since a large realised error can receive a small or zero weight.

D. Analysis Protocol

Unless stated otherwise, analyses use the replicate means $y_m(o,a)$ and are descriptive within the evaluated task set: the descriptor and the weighting directions were informed by these outcomes, correlations are reported without significance claims, and residualisation removes additive group differences without supporting causal interpretation.

Decomposition. For each metric, the total centred sum of squares over all runs is split into order, algorithm, order–algorithm interaction, and within-cell training components (Appendix A-C). The order component is further split into starting-task identity and order within starting task, and the decomposition is repeated for the 18 orders that do not start with task A. Replicate labels are local to each order–algorithm cell and do not imply matched initial states across cells.

Warm-start error. To separate the first stage, which involves no previous task, from the continual part of a run, the current-task error is also reported without its first term (Section V-C).

Associations. Spearman correlations with the 16 algorithm–metric series use average ranks for ties. Starting-task dependence is assessed by excluding A-first orders and by correlating ranked variables after removing starting-task group means; the same residualisation with position-of-A group means isolates what D adds beyond the position of task A.

Replicate bootstrap. The three replicates of every order–algorithm cell are resampled with replacement (5,000 draws, fixed seed), and the order means, policy scores, and per-order leaders are recomputed. This yields the probability that each algorithm attains the smallest policy score and an interval for its number of per-order wins. A per-order leader is called separable when its gap to the runner-up exceeds the pooled sample standard deviation of the two algorithms. These quantities describe sensitivity to training variation only, since the 24 orders are the complete population of this task set. With three replicates, plug-in resampling underestimates the variance of a cell mean by the factor $(S-1)/S$, so the reported probabilities slightly overstate how decisive a leader is; rescaling the resampled deviations by $\sqrt{S/(S-1)}$ changes them by at most 0.05.

Paths. We also compare the orders with the largest and smallest total transition distance,

$$L(o) = \sum_{r=1}^{N-1} d_{o_r, o_{r+1}}, \tag{15}$$

within each fixed starting task. Path length and slope are different summaries: L uses all three transitions, whereas D omits the middle one.

V. Experimental Setup

A. Motion Forecasting Tasks and Backbone

Each forecasting task comprises trajectory records from one category of road layout in INTERACTION [40]. Twelve location subsets are grouped into four tasks: *A, Urban Merge*, lane convergence and zipper merging at the German MT site; *B, Highway Merge*, ramp entry and lane convergence at the Chinese ZS highway site (archived subsets ZS0 and ZS2); *C, Intersection*, crossing and turning interactions at the US subsets MA, GL, EP0, and EP1; and *D, Roundabout*, entry, circulation, and exit at EP, SR, and FT in the US, OF in Germany, and LN in China. The task arrays contain 5,378, 50,905, 134,269, and 103,540 trajectory records, respectively; these are stored records, not independent observations. Task A is thus about 10 to 25 times smaller than the other tasks, which matters for the first-stage results below.

Fig. 3 compares the locations and spreads of the empirical target-agent feature distributions. Urban merging has the highest median speed (6.29 m/s), whereas highway merging combines a lower median (2.87 m/s), reflecting queuing on the ramp, with the largest 95th percentile (17.56 m/s). Intersections have the lowest median speed (2.27 m/s), and intersections and roundabouts have the wider acceleration interquartile ranges.

The forecasting input is $\mathbf{X} = (\mathbf{X}_{\mathrm{TA}}, \mathbf{X}_{\mathrm{SA}}, \mathbf{M})$, containing the target-agent history, surrounding-agent histories, and map context. The ground-truth output $\mathbf{Y} \in \mathbb{R}^2$ is the target endpoint. UQnet [17] combines agent and map encoders with a spatial prediction head to produce a heatmap over future endpoints, from which the predictor samples W candidate endpoints $\{\hat{\mathbf{Y}}_w\}_{w=1}^{W}$. The training loss follows the heatmap-based forecasting formulation in the backbone work.

Fig. 4 details the encoder and prediction branches. Lane attention encodes the map into $\mathbf{H}_{\mathrm{M}}$, a multilayer perceptron embeds spatial coordinates as $\mathbf{H}_{\mathrm{C}}$, and a trajectory subnet extracts target-vehicle and surrounding-agent features, $\mathbf{H}_{\mathrm{TV}}$ and $\mathbf{H}_{\mathrm{S}}$. The interaction module fuses vehicle and map features, which the decoder combines with coordinate features to predict an endpoint heatmap. The regularisation network applies two cross-attention blocks to map, target-vehicle, and

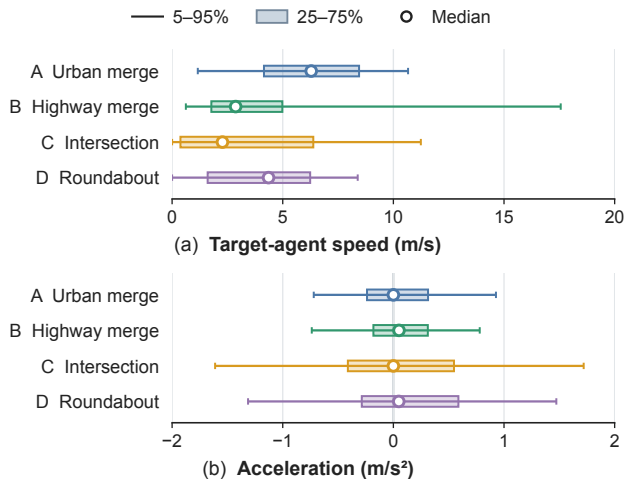


Fig. 3. Task feature distributions. Dots, bands, and whiskers indicate medians, interquartile ranges, and 5th–95th percentiles. Acceleration is restricted to $[-5, 5]$ m/s^2.

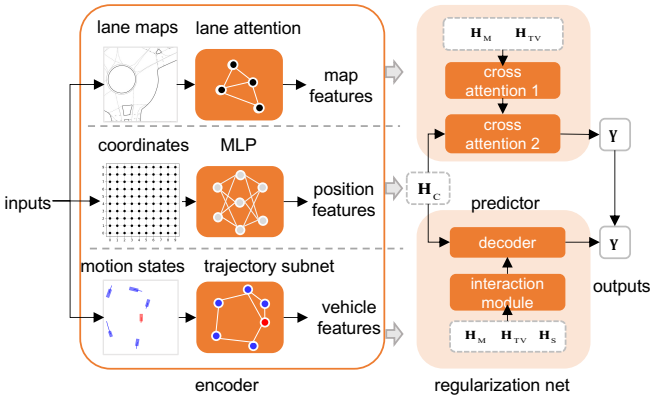


Fig. 4. UQnet backbone: feature encoding, interaction modelling, endpoint prediction, and the regularisation network used during training (adapted from [17]).

coordinate features during training and is removed at inference [17]. All compared algorithms use the same predictor; the evaluation concerns endpoint prediction only.

B. Algorithms and Experimental Coverage

We compare four algorithms under the common prediction architecture (Table I). *Vanilla* optimises the current task without memory and provides the reference for unconstrained adaptation. *DER* (dark experience replay) [8] rehearses stored inputs with output matching to preserve previous predictions, in a forecasting adaptation of the original classification method. *SyReM* (synergetic memory rehearsal) [15] combines gradient-based rehearsal selection with an average-memory loss constraint. *GSM* [14] is the equal-allocation variant of dynamic gradient scenario memory (D-GSM), with scenario-organised memory (a scenario is one of the four tasks here) and gradient constraints derived from previous scenarios. The selection spans output distillation, selective constrained rehearsal, and scenario-wise constraints; all memory-based

TABLE I
PUBLISHED MECHANISMS OF THE COMPARED ALGORITHMS. STORED EXAMPLES SERVE OUTPUT MATCHING, REHEARSAL, OR GRADIENT CONSTRAINTS; SCENARIO PARTITION ALLOCATES MEMORY BY PAST TASK.

Algorithm	Stored examples	Output matching	Scenario partition	Gradient constraint	Gradient-based selection
Vanilla	×	×	×	×	×
DER	✓	✓	×	×	×
SyReM	✓	×	×	✓	✓
GSM	✓	×	✓	✓	×

runs use the recorded capacity of 2,000. Training configurations are maintained separately from the released analysis (Appendix A-D).

The experiment evaluates all $4! = 24$ orders of $N = 4$ tasks, with four algorithms and three training replicates per combination (288 runs). Testing every task after each stage yields 1,152 stage logs and 4,608 stage–task records. Evaluation data and analysis code are available online[1]; resources for model training are maintained separately.

C. Forecasting and Continual Learning Metrics

Minimum final displacement error (FDE) measures the distance from the ground-truth endpoint to the closest predicted candidate:

$$\mathrm{FDE} = \min_{1 \le w \le W} \|\hat{\mathbf{Y}}_w - \mathbf{Y}\|_2. \tag{16}$$

Miss rate (MR) reports the percentage of evaluated cases whose closest candidate lies beyond the evaluator's endpoint tolerance. We use the saved FDE and MR values; the present analysis does not recompute prediction-level metrics.

For either error metric, let $E_{r,j}^{(o,a,s)}$ denote the result on task identity j after stage r. We reorder columns by presentation position and define

$$R_{r,i}^{(o,a,s)} = E_{r,o_i}^{(o,a,s)}. \tag{17}$$

Thus $R_{r,r}$ always evaluates the task just learned, even when its identity is not r. Suppressing the superscripts, the summaries are

$$\mathrm{CT} = \frac{1}{N} \sum_{r=1}^{N} R_{r,r}, \tag{18}$$

$$\mathrm{BWT} = \frac{1}{N-1} \sum_{r=2}^{N} \frac{1}{r-1} \sum_{i=1}^{r-1} (R_{r,i} - R_{i,i}), \tag{19}$$

$$\mathrm{Final} = \frac{1}{N} \sum_{i=1}^{N} R_{N,i}. \tag{20}$$

The current-task error CT measures error upon initial learning of each task. The backward transfer BWT averages subsequent error changes over stages: positive values indicate degradation and negative values improvement relative to the initial fit, reversing the accuracy-based sign convention. FDE is in metres, MR in percent, and MR-BWT in percentage points. Lower CT and Final indicate better absolute predictions; lower BWT

[1]https://github.com/lzrbit/task_order

TABLE II
PERCENTAGE OF THE TOTAL CENTRED SUM OF SQUARES PER COMPONENT. THE ORDER TERM IS SPLIT INTO STARTING-TASK IDENTITY (START) AND ORDER WITHIN STARTING TASK; THE LAST THREE COLUMNS GIVE THE 18 ORDERS THAT DO NOT START WITH A.

	All 24 orders (%)					Excl. A-first (%)		
Metric	Start	Order| start	Alg.	Order× alg.	Within	Order	Alg.	Within
FDE-CT	69.6	1.7	1.3	11.3	16.0	30.6	2.3	50.8
MR-CT	58.3	6.3	1.7	10.3	23.4	24.5	2.4	51.2
FDE-BWT	75.7	0.5	0.9	9.7	13.1	19.3	17.4	45.9
MR-BWT	89.8	1.5	3.4	1.3	4.0	39.5	26.5	25.6
FDE-$\mathrm{CT}_{2:N}$	4.9	20.7	2.3	20.2	52.0	24.6	1.8	53.1
MR-$\mathrm{CT}_{2:N}$	7.9	15.9	2.8	24.3	49.2	21.3	2.1	51.2

indicates greater relative improvement and need not imply lower final error.

The first term of CT, $R_{1,1}$, is the error of single-task learning from the initial state and involves no previous task. We therefore also report the warm-start current-task error

$$\mathrm{CT}_{2:N} = \frac{1}{N-1}\sum_{r=2}^{N} R_{r,r}, \tag{21}$$

so that $\mathrm{CT} = [R_{1,1} + (N-1)\,\mathrm{CT}_{2:N}]/N$. We write FDE-CT, MR-BWT, and so on for the metric–summary combinations.

VI. EXPERIMENTAL RESULTS

A. Order-Resolved Algorithm Comparisons

Tables III and IV report all orders. Each entry is a mean and sample standard deviation across three replicates, and bold marks the per-order minima that are separable from the runner-up.

Table II decomposes the total centred sum of squares. Task order accounts for 71.3%, 64.6%, 76.3%, and 91.3% in FDE-CT, MR-CT, FDE-BWT, and MR-BWT, against 1.3%, 1.7%, 0.9%, and 3.4% for the algorithm and 11.3%, 10.3%, 9.7%, and 1.3% for their interaction. Starting-task identity alone accounts for 69.6%, 58.3%, 75.7%, and 89.8%, which is 90% to 99% of the order term. Among the 18 orders that do not start with A, the order term falls to 30.6%, 24.5%, 19.3%, and 39.5%; within-cell training variation becomes the largest component for FDE-CT, MR-CT, and FDE-BWT (50.8%, 51.2%, and 45.9%), whereas for MR-BWT the order term remains largest and the algorithm term (26.5%) is comparable to within-cell variation (25.6%). Shares are not adjusted for degrees of freedom: under pure within-cell noise the interaction would take 5.8%, 8.4%, 4.7%, and 1.4% (Appendix A-C), so it exceeds this floor clearly only for FDE-CT and FDE-BWT.

For FDE-CT, the algorithm-averaged order means range from 0.97 to 6.92 m, a spread of 5.95 m. The median gap between the best and worst algorithm within an order is 0.87 m, and the gap between the largest and smallest uniform-order algorithm means is 0.62 m. Changing the order can therefore have a larger numerical effect than changing the algorithm, but most of that order effect is the identity of the first task.

Every algorithm has the smallest replicate-mean FDE-CT on at least three orders: Vanilla on 3, DER on 9, SyReM on 9, and GSM on 3, with bootstrap intervals of $[1, 8]$, $[5, 12]$, $[5, 12]$, and $[1, 6]$. All four also lead on at least one MR-CT order (2, 8, 11, and 3), whereas DER never attains a per-order minimum of either BWT metric. The per-order leader is separable from the runner-up in only 4, 2, 10, and 11 of the 24 orders for FDE-CT, MR-CT, FDE-BWT, and MR-BWT, and the separable FDE-BWT and MR-BWT leaders are nearly all SyReM (9 of 10 and 10 of 11). Most reversals of replicate-mean ranks therefore lie within training variability. Fig. 1 shows one that does not: on A–C–B–D, SyReM leads with 3.31 m, whereas on A–C–D–B, which swaps the last two tasks, DER leads with 3.48 m; the leaders' gaps to the runner-up are 3.65 and 1.10 pooled standard deviations. Both orders start with A, and Fig. 1(b) shows that each leader owes its advantage to a lower first-stage error on A (10.7 m against 21.5 to 24.8 m, and 10.4 m against 16.0 to 34.3 m), whereas the warm-start errors of the four algorithms lie within 0.9 to 1.5 m and no algorithm is separable; the reversal is a first-stage effect, which the next subsection examines.

B. The Starting Task and the Cold Start of Task A

Table V resolves the FDE summaries by starting task. When task A is learned first, the first-stage error $R_{1,1}$ averages 21.52 m over the 72 runs, ranging from 4.7 to 42.9 m, with a first-stage miss rate of 93.8%; the other starting tasks give 1.04 to 2.77 m. When A is learned at positions 2, 3, and 4 instead, its own stage error is 3.30, 1.87, and 1.18 m, against 0.88 to 1.32 m for the other tasks at warm positions, and its final error is 0.96 to 1.18 m in every position. The algorithm-averaged FDE-CT accordingly falls from 6.25 m with A in position 1 to 1.76, 1.42, and 1.20 m with A in positions 2, 3, and 4. The first stage is single-task learning from the initial state, so no cross-task retention mechanism is involved; we attribute the failure to the sample size of task A under the common training schedule, although A is also the most distinct task in speed and a single site, so sample size and distribution cannot be separated with these runs. The first stage is not identical across algorithms: GSM reproduces Vanilla's first-stage error under the shared-state replicate label, whereas DER and SyReM reach lower A-first first-stage errors (17.6 and 18.5 m against 26.4 and 23.6 m), consistent with their within-task losses acting while the memory fills (Appendix A-D).

The warm-start error $\mathrm{CT}_{2:N}$ removes the first-stage term. Its order component drops to 25.6% (FDE) and 23.8% (MR), the starting-task share to 4.9% and 7.9%, and within-cell training variation rises to 52.0% and 49.2% (Table II). A-first orders are then no longer the worst starting group (1.16 m and 9.05% against 1.15 to 1.50 m and 9.86% to 14.34% for the other starting tasks), but this comparison is not like for like: $\mathrm{CT}_{2:N}$ for A-first orders averages tasks B, C, and D only, whereas for the other starting tasks it includes the warm-position error of A, which is about twice that of the other tasks. The residual order effect is of the same magnitude as training variation, yet it remains structured by the position of A, whose group means account for 13.3% (FDE) and 15.7% (MR) of the total sum of squares, half to two-thirds of the residual order term.

TABLE III
CURRENT-TASK ERRORS BY TASK ORDER (MEAN ± SAMPLE STANDARD DEVIATION OVER THREE REPLICATES). FDE IN METRES; MR IN PERCENT. BOLD: PER-ORDER MINIMUM SEPARABLE FROM THE RUNNER-UP (SECTION IV-D).

Order	Vanilla		DER		SyReM		GSM	
	FDE-CT	MR-CT	FDE-CT	MR-CT	FDE-CT	MR-CT	FDE-CT	MR-CT
ABCD	$8.25_{\pm 0.46}$	$32.83_{\pm 2.59}$	$6.17_{\pm 2.13}$	$31.75_{\pm 1.97}$	$7.17_{\pm 3.71}$	$31.74_{\pm 1.65}$	$6.08_{\pm 1.36}$	$31.84_{\pm 0.56}$
ABDC	$4.98_{\pm 2.49}$	$30.83_{\pm 4.43}$	$6.67_{\pm 2.39}$	$32.20_{\pm 1.14}$	$6.55_{\pm 2.48}$	$28.76_{\pm 2.27}$	$8.00_{\pm 3.49}$	$34.38_{\pm 0.52}$
ACBD	$7.33_{\pm 3.01}$	$33.28_{\pm 4.02}$	$6.69_{\pm 2.57}$	$29.44_{\pm 1.99}$	$\mathbf{3.31_{\pm 0.31}}$	$\mathbf{25.33_{\pm 0.67}}$	$6.23_{\pm 1.09}$	$31.86_{\pm 2.69}$
ACDB	$9.26_{\pm 1.95}$	$29.96_{\pm 1.81}$	$\mathbf{3.48_{\pm 1.02}}$	$28.85_{\pm 3.15}$	$4.71_{\pm 1.20}$	$27.92_{\pm 4.05}$	$5.00_{\pm 1.84}$	$30.21_{\pm 1.17}$
ADBC	$9.20_{\pm 1.98}$	$31.51_{\pm 2.50}$	$3.38_{\pm 0.70}$	$28.18_{\pm 4.41}$	$3.90_{\pm 1.97}$	$26.08_{\pm 6.67}$	$9.17_{\pm 2.11}$	$35.83_{\pm 0.44}$
ADCB	$5.92_{\pm 2.54}$	$26.34_{\pm 4.40}$	$5.10_{\pm 3.73}$	$25.08_{\pm 1.65}$	$6.53_{\pm 2.44}$	$29.38_{\pm 0.54}$	$6.97_{\pm 0.34}$	$32.37_{\pm 2.82}$
BACD	$2.40_{\pm 1.13}$	$21.10_{\pm 8.23}$	$2.97_{\pm 1.77}$	$21.92_{\pm 13.62}$	$2.16_{\pm 0.20}$	$21.02_{\pm 3.63}$	$2.66_{\pm 1.38}$	$21.34_{\pm 10.22}$
BADC	$2.13_{\pm 0.63}$	$20.75_{\pm 6.81}$	$2.09_{\pm 0.65}$	$19.11_{\pm 8.98}$	$1.61_{\pm 0.48}$	$15.98_{\pm 5.36}$	$2.70_{\pm 1.27}$	$23.88_{\pm 7.66}$
BCAD	$1.45_{\pm 0.37}$	$12.31_{\pm 4.21}$	$2.14_{\pm 1.02}$	$22.57_{\pm 13.92}$	$1.21_{\pm 0.55}$	$8.46_{\pm 4.87}$	$1.52_{\pm 0.32}$	$13.24_{\pm 4.24}$
BCDA	$1.21_{\pm 0.32}$	$9.16_{\pm 1.51}$	$1.12_{\pm 0.30}$	$7.34_{\pm 2.98}$	$1.25_{\pm 0.22}$	$10.65_{\pm 2.05}$	$1.72_{\pm 0.57}$	$16.51_{\pm 9.91}$
BDAC	$1.62_{\pm 0.19}$	$15.08_{\pm 3.63}$	$1.75_{\pm 0.41}$	$13.61_{\pm 1.82}$	$1.48_{\pm 0.28}$	$15.67_{\pm 7.44}$	$1.80_{\pm 0.50}$	$16.02_{\pm 7.99}$
BDCA	$1.53_{\pm 0.53}$	$11.71_{\pm 3.11}$	$1.33_{\pm 0.54}$	$12.87_{\pm 7.73}$	$1.63_{\pm 0.48}$	$12.85_{\pm 3.69}$	$1.84_{\pm 0.19}$	$17.31_{\pm 3.29}$
CABD	$1.27_{\pm 0.28}$	$14.74_{\pm 5.90}$	$1.13_{\pm 0.25}$	$10.49_{\pm 4.61}$	$1.28_{\pm 0.48}$	$14.34_{\pm 9.44}$	$1.01_{\pm 0.20}$	$7.81_{\pm 3.57}$
CADB	$1.66_{\pm 0.42}$	$22.08_{\pm 6.90}$	$1.07_{\pm 0.35}$	$9.75_{\pm 7.29}$	$1.15_{\pm 0.23}$	$11.32_{\pm 5.51}$	$0.97_{\pm 0.15}$	$7.47_{\pm 3.60}$
CBAD	$1.07_{\pm 0.10}$	$9.64_{\pm 1.13}$	$1.07_{\pm 0.30}$	$6.78_{\pm 2.11}$	$0.94_{\pm 0.08}$	$7.34_{\pm 2.35}$	$1.58_{\pm 0.36}$	$17.08_{\pm 8.05}$
CBDA	$1.02_{\pm 0.11}$	$8.37_{\pm 1.34}$	$1.12_{\pm 0.37}$	$11.67_{\pm 8.13}$	$1.00_{\pm 0.15}$	$8.76_{\pm 3.74}$	$1.05_{\pm 0.26}$	$7.85_{\pm 3.42}$
CDAB	$1.32_{\pm 0.79}$	$12.34_{\pm 10.76}$	$\mathbf{0.91_{\pm 0.13}}$	$6.68_{\pm 3.23}$	$1.22_{\pm 0.63}$	$9.62_{\pm 8.52}$	$1.16_{\pm 0.20}$	$11.07_{\pm 2.88}$
CDBA	$1.05_{\pm 0.34}$	$7.88_{\pm 3.92}$	$0.95_{\pm 0.03}$	$7.58_{\pm 0.53}$	$0.89_{\pm 0.09}$	$\mathbf{5.91_{\pm 2.21}}$	$0.99_{\pm 0.18}$	$8.20_{\pm 4.24}$
DABC	$1.76_{\pm 0.93}$	$18.29_{\pm 13.10}$	$\mathbf{1.08_{\pm 0.20}}$	$10.46_{\pm 4.87}$	$1.84_{\pm 1.25}$	$17.54_{\pm 11.58}$	$2.04_{\pm 0.82}$	$21.88_{\pm 10.08}$
DACB	$1.75_{\pm 0.12}$	$23.72_{\pm 5.06}$	$1.14_{\pm 0.17}$	$13.03_{\pm 4.88}$	$1.37_{\pm 0.59}$	$12.03_{\pm 4.42}$	$2.92_{\pm 2.14}$	$23.69_{\pm 7.29}$
DBAC	$1.27_{\pm 0.50}$	$9.68_{\pm 3.12}$	$2.39_{\pm 1.76}$	$21.68_{\pm 11.80}$	$1.36_{\pm 0.53}$	$14.11_{\pm 9.07}$	$1.49_{\pm 0.56}$	$19.62_{\pm 11.26}$
DBCA	$1.14_{\pm 0.18}$	$13.15_{\pm 3.24}$	$1.10_{\pm 0.16}$	$10.39_{\pm 3.88}$	$1.18_{\pm 0.23}$	$11.06_{\pm 1.64}$	$1.36_{\pm 0.45}$	$15.95_{\pm 7.94}$
DCAB	$1.04_{\pm 0.06}$	$10.12_{\pm 1.57}$	$2.24_{\pm 0.90}$	$24.82_{\pm 7.94}$	$1.01_{\pm 0.13}$	$7.89_{\pm 2.05}$	$1.02_{\pm 0.11}$	$9.98_{\pm 2.72}$
DCBA	$1.01_{\pm 0.08}$	$9.44_{\pm 2.96}$	$1.03_{\pm 0.13}$	$9.01_{\pm 2.85}$	$1.06_{\pm 0.10}$	$11.17_{\pm 2.48}$	$1.32_{\pm 0.52}$	$14.56_{\pm 7.85}$

TABLE IV
BACKWARD TRANSFER BY TASK ORDER (MEAN ± SAMPLE STANDARD DEVIATION OVER THREE REPLICATES). FDE IN METRES; MR IN PERCENTAGE POINTS. BOLD AS IN TABLE III.

Order	Vanilla		DER		SyReM		GSM	
	FDE-BWT	MR-BWT	FDE-BWT	MR-BWT	FDE-BWT	MR-BWT	FDE-BWT	MR-BWT
ABCD	$-16.39_{\pm 1.57}$	$-46.19_{\pm 3.15}$	$-10.99_{\pm 5.05}$	$-41.41_{\pm 0.95}$	$-14.77_{\pm 8.97}$	$-52.95_{\pm 2.18}$	$-12.00_{\pm 3.63}$	$-54.18_{\pm 2.86}$
ABDC	$-8.41_{\pm 5.26}$	$-42.47_{\pm 2.39}$	$-12.28_{\pm 5.50}$	$-44.11_{\pm 2.93}$	$-13.55_{\pm 5.89}$	$-54.58_{\pm 3.96}$	$-15.61_{\pm 9.07}$	$-48.23_{\pm 11.43}$
ACBD	$-13.97_{\pm 7.56}$	$-47.29_{\pm 3.65}$	$-13.49_{\pm 6.02}$	$-46.46_{\pm 2.77}$	$-5.96_{\pm 0.70}$	$-50.20_{\pm 1.11}$	$-12.35_{\pm 2.34}$	$-49.80_{\pm 3.44}$
ACDB	$\mathbf{-19.84_{\pm 4.77}}$	$-42.47_{\pm 11.19}$	$-5.36_{\pm 2.77}$	$-37.45_{\pm 12.64}$	$-9.18_{\pm 3.12}$	$-53.14_{\pm 5.23}$	$-8.97_{\pm 5.24}$	$-48.22_{\pm 7.97}$
ADBC	$-19.42_{\pm 4.94}$	$-45.86_{\pm 2.94}$	$-5.44_{\pm 1.35}$	$-40.02_{\pm 8.01}$	$-7.15_{\pm 4.53}$	$-45.12_{\pm 8.76}$	$-18.75_{\pm 5.52}$	$\mathbf{-50.08_{\pm 2.73}}$
ADCB	$-11.50_{\pm 6.68}$	$-33.69_{\pm 21.46}$	$-10.01_{\pm 9.22}$	$-41.18_{\pm 7.96}$	$-13.73_{\pm 5.86}$	$\mathbf{-56.97_{\pm 0.56}}$	$-14.32_{\pm 0.59}$	$-49.42_{\pm 2.57}$
BACD	$0.38_{\pm 1.30}$	$-4.77_{\pm 2.03}$	$0.38_{\pm 0.41}$	$2.80_{\pm 7.91}$	$-0.35_{\pm 0.64}$	$-12.22_{\pm 2.50}$	$-0.68_{\pm 1.66}$	$-8.45_{\pm 10.69}$
BADC	$-0.95_{\pm 0.77}$	$-8.34_{\pm 10.34}$	$-0.13_{\pm 0.46}$	$-3.38_{\pm 7.23}$	$-0.18_{\pm 1.07}$	$-7.88_{\pm 12.83}$	$-0.36_{\pm 1.37}$	$-5.76_{\pm 7.80}$
BCAD	$0.17_{\pm 0.84}$	$4.50_{\pm 2.97}$	$0.57_{\pm 0.80}$	$7.14_{\pm 4.99}$	$-0.51_{\pm 1.00}$	$-4.43_{\pm 9.09}$	$-0.63_{\pm 0.44}$	$-3.01_{\pm 2.01}$
BCDA	$0.97_{\pm 1.18}$	$12.43_{\pm 9.35}$	$1.08_{\pm 0.29}$	$10.74_{\pm 3.78}$	$-0.47_{\pm 0.65}$	$-3.43_{\pm 7.03}$	$-0.72_{\pm 0.92}$	$1.58_{\pm 3.47}$
BDAC	$0.44_{\pm 0.85}$	$4.86_{\pm 1.12}$	$-0.01_{\pm 0.80}$	$5.39_{\pm 4.17}$	$-0.38_{\pm 0.66}$	$-5.72_{\pm 3.49}$	$-0.85_{\pm 0.32}$	$-2.73_{\pm 2.70}$
BDCA	$0.43_{\pm 1.25}$	$7.10_{\pm 11.32}$	$0.87_{\pm 1.00}$	$11.62_{\pm 2.77}$	$-1.02_{\pm 1.18}$	$-0.11_{\pm 3.79}$	$-0.81_{\pm 0.70}$	$1.27_{\pm 4.80}$
CABD	$0.66_{\pm 0.14}$	$6.36_{\pm 4.90}$	$0.68_{\pm 0.13}$	$9.91_{\pm 3.58}$	$\mathbf{-0.10_{\pm 0.38}}$	$\mathbf{-2.38_{\pm 7.88}}$	$0.22_{\pm 0.05}$	$5.57_{\pm 1.98}$
CADB	$0.38_{\pm 0.15}$	$3.60_{\pm 5.30}$	$0.45_{\pm 0.14}$	$8.91_{\pm 3.94}$	$\mathbf{0.05_{\pm 0.12}}$	$1.92_{\pm 3.45}$	$0.42_{\pm 0.30}$	$8.87_{\pm 6.16}$
CBAD	$0.97_{\pm 0.31}$	$11.47_{\pm 3.74}$	$0.80_{\pm 0.19}$	$14.11_{\pm 1.94}$	$0.11_{\pm 0.17}$	$\mathbf{0.22_{\pm 4.51}}$	$0.46_{\pm 0.63}$	$10.30_{\pm 4.48}$
CBDA	$1.12_{\pm 0.19}$	$13.35_{\pm 3.41}$	$1.29_{\pm 0.34}$	$15.87_{\pm 2.03}$	$\mathbf{-0.07_{\pm 0.25}}$	$\mathbf{-2.14_{\pm 6.78}}$	$0.26_{\pm 0.37}$	$7.52_{\pm 6.05}$
CDAB	$0.64_{\pm 0.24}$	$14.77_{\pm 3.67}$	$0.66_{\pm 0.18}$	$14.69_{\pm 2.58}$	$\mathbf{0.21_{\pm 0.25}}$	$\mathbf{3.29_{\pm 3.01}}$	$0.58_{\pm 0.13}$	$13.43_{\pm 1.53}$
CDBA	$0.82_{\pm 0.12}$	$16.91_{\pm 1.55}$	$0.73_{\pm 0.48}$	$13.16_{\pm 8.55}$	$\mathbf{0.21_{\pm 0.15}}$	$\mathbf{4.24_{\pm 3.76}}$	$0.85_{\pm 0.51}$	$15.54_{\pm 6.66}$
DABC	$0.40_{\pm 0.45}$	$10.65_{\pm 7.35}$	$0.71_{\pm 0.16}$	$14.06_{\pm 2.05}$	$-0.49_{\pm 1.34}$	$\mathbf{-3.59_{\pm 12.00}}$	$1.38_{\pm 1.66}$	$15.06_{\pm 9.24}$
DACB	$0.45_{\pm 0.07}$	$5.37_{\pm 3.07}$	$0.64_{\pm 0.18}$	$11.91_{\pm 4.44}$	$0.29_{\pm 0.38}$	$2.62_{\pm 1.80}$	$-0.75_{\pm 1.53}$	$0.84_{\pm 2.56}$
DBAC	$0.86_{\pm 0.49}$	$17.83_{\pm 2.77}$	$1.04_{\pm 0.34}$	$15.53_{\pm 2.89}$	$\mathbf{0.21_{\pm 0.06}}$	$\mathbf{3.21_{\pm 1.64}}$	$0.90_{\pm 0.16}$	$8.44_{\pm 5.08}$
DBCA	$1.03_{\pm 0.17}$	$17.66_{\pm 3.60}$	$0.95_{\pm 0.14}$	$16.63_{\pm 3.17}$	$\mathbf{0.01_{\pm 0.19}}$	$\mathbf{2.55_{\pm 1.02}}$	$0.84_{\pm 0.58}$	$13.79_{\pm 5.05}$
DCAB	$0.73_{\pm 0.11}$	$15.61_{\pm 1.74}$	$0.73_{\pm 0.37}$	$12.99_{\pm 5.99}$	$\mathbf{0.21_{\pm 0.22}}$	$6.13_{\pm 6.09}$	$0.54_{\pm 0.25}$	$10.47_{\pm 4.67}$
DCBA	$0.85_{\pm 0.32}$	$15.16_{\pm 6.63}$	$0.91_{\pm 0.27}$	$18.19_{\pm 3.59}$	$\mathbf{0.03_{\pm 0.12}}$	$\mathbf{-0.22_{\pm 4.14}}$	$0.53_{\pm 0.41}$	$11.18_{\pm 5.78}$

TABLE V
FDE (M) BY STARTING TASK OVER 72 RUNS PER ROW: INITIAL ERROR $R_{1,1}$, CT WITH AND WITHOUT ($\mathrm{CT}_{2:N}$) THE FIRST STAGE, SUBSEQUENT ERROR P, REFERENCE B, $\mathrm{BWT} = P - B$, AND FINAL ERROR.

Start	$R_{1,1}$	CT	$\mathrm{CT}_{2:N}$	P	B	BWT	Final
A	21.52	6.25	1.16	1.46	13.69	−12.23	1.28
B	2.77	1.81	1.48	2.34	2.46	−0.12	1.52
C	1.04	1.12	1.15	1.64	1.13	0.52	1.50
D	1.32	1.45	1.50	2.04	1.50	0.54	1.48

C. Descriptor Associations and the Position of Task A

Table VI gives the associations of the five candidate task-distance matrices with CL performance, with tied ranks preserved. MMD and W1 on speed induce the same ordering of descriptor values and hence identical Spearman correlations, although their magnitudes and resulting weights differ; the primary analysis retains the interpretable W1 scale. Under the acceleration distance, the pair containing A is the closer pair in two of the three partitions, so D changes sign for the 16 orders built on those partitions and no longer tracks the position of A (Spearman -0.61 against $+0.79$ for speed); the correlations in the last column reverse in sign accordingly, with smaller

TABLE VI
SPEARMAN CORRELATIONS BETWEEN CANDIDATE TRANSITION-DIRECTION DESCRIPTORS AND ORDER-LEVEL PERFORMANCE. W1: WASSERSTEIN-1. BOLD: $|\rho| \geq 0.7$.

Algorithm	Metric	W1 speed	KL speed	JS speed	MMD speed	W1 accel.
Vanilla	FDE-CT	$\mathbf{-0.75}$	-0.23	-0.20	$\mathbf{-0.75}$	$+0.69$
	MR-CT	$\mathbf{-0.75}$	-0.09	-0.12	$\mathbf{-0.75}$	$+0.69$
	FDE-BWT	$+0.66$	$+0.13$	$+0.10$	$+0.66$	-0.55
	MR-BWT	$\mathbf{+0.81}$	$+0.26$	$+0.32$	$\mathbf{+0.81}$	-0.46
DER	FDE-CT	-0.57	-0.29	-0.36	-0.57	$+0.46$
	MR-CT	-0.54	-0.22	-0.27	-0.54	$+0.48$
	FDE-BWT	$+0.67$	$+0.10$	$+0.12$	$+0.67$	-0.52
	MR-BWT	$\mathbf{+0.74}$	$+0.28$	$+0.32$	$\mathbf{+0.74}$	-0.43
SyReM	FDE-CT	-0.67	-0.29	-0.26	-0.67	$\mathbf{+0.74}$
	MR-CT	$\mathbf{-0.72}$	-0.22	-0.23	$\mathbf{-0.72}$	$\mathbf{+0.76}$
	FDE-BWT	$+0.55$	$+0.31$	$+0.36$	$+0.55$	-0.35
	MR-BWT	$\mathbf{+0.72}$	$+0.35$	$+0.43$	$\mathbf{+0.72}$	-0.42
GSM	FDE-CT	-0.57	-0.30	-0.29	-0.57	$+0.66$
	MR-CT	-0.53	-0.23	-0.23	-0.53	$+0.69$
	FDE-BWT	$+0.55$	$+0.32$	$+0.30$	$+0.55$	-0.37
	MR-BWT	$+0.69$	$+0.40$	$+0.44$	$+0.69$	-0.43
Mean $\lvert\rho\rvert$		0.66	0.25	0.27	0.66	0.54

TABLE VII
MEAN SPEARMAN ASSOCIATIONS ACROSS ALGORITHMS. $\rho(D)$: DESCRIPTOR AND ORDER MEANS. EXCL. A-FIRST: 18 ORDERS. START-ADJ.: RANK RESIDUALS AFTER REMOVING STARTING-TASK MEANS. $\rho(p_A)$: POSITION OF TASK A. p_A-ADJ.: RANK RESIDUALS AFTER REMOVING POSITION-OF-A MEANS.

Metric	$\rho(D)$	Excl. A-first	Start-adj.	$\rho(p_A)$	p_A-adj.
FDE-CT	-0.64	-0.47	-0.48	-0.77	-0.23
MR-CT	-0.63	-0.43	-0.49	-0.79	-0.11
FDE-BWT	$+0.61$	$+0.40$	$+0.29$	$+0.73$	$+0.22$
MR-BWT	$+0.74$	$+0.64$	$+0.62$	$+0.77$	$+0.53$
FDE-CT$_{2:N}$	-0.37	-0.49	-0.58	-0.20	-0.27
MR-CT$_{2:N}$	-0.27	-0.47	-0.50	-0.10	-0.20

TABLE VIII
FINAL ERRORS: MEAN [MINIMUM, MAXIMUM] OVER ORDER-LEVEL REPLICATE MEANS, AND THE BOOTSTRAP PROBABILITY (PROB.) OF THE SMALLEST UNIFORM MEAN. FDE IN METRES; MR IN PERCENT. BOLD: SMALLEST MEAN.

Algorithm	FDE: mean [min, max]	Prob.	MR: mean [min, max]	Prob.
Vanilla	1.63 [1.18, 2.14]	0.00	17.22 [10.84, 25.67]	0.00
DER	1.63 [1.17, 2.34]	0.00	17.80 [11.47, 29.09]	0.00
SyReM	**1.08 [0.87, 1.99]**	1.00	**9.23 [4.38, 24.69]**	1.00
GSM	1.43 [1.03, 2.44]	0.00	15.86 [8.05, 36.25]	0.00

magnitudes for BWT.

Across the 16 algorithm–metric series, absolute Spearman correlations with D range from 0.53 to 0.81, with algorithm-averaged values of -0.64, -0.63, $+0.61$, and $+0.74$ for FDE-CT, MR-CT, FDE-BWT, and MR-BWT. Fig. 5 resolves these associations for each algorithm and metric across the six tied descriptor levels; the two clusters visible in every panel coincide with A in positions 1–2 (negative D) and 3–4 (positive D).

Table VII quantifies this reading. The position of A alone correlates more strongly than D with each of the four primary metrics (-0.77, -0.79, $+0.73$, and $+0.77$). Excluding A-first orders reduces the mean association with D in each of these metrics, removing starting-task means leaves -0.48, -0.49, $+0.29$, and $+0.62$, and removing position-of-A means leaves -0.23, -0.11, $+0.22$, and $+0.53$. For the warm-start error the marginal associations are weaker (-0.37 and -0.27 with D; -0.20 and -0.10 with the position of A), and excluding A-first orders strengthens rather than weakens them, because A-first orders combine the most negative D with low warm-start error. On this task set, D therefore mainly indexes where the most distinct and smallest task is placed; what it adds beyond that position is small for the CT metrics and moderate only for MR-BWT. A sensitivity comparison with symmetrised historical W1 estimates produces the same descriptor ranking (Appendix A-A).

D. Backward Transfer and the Initial Reference

The large negative BWT values for A-first orders motivate a decomposition of the initial reference. Expanding (19) gives

$$\mathrm{BWT} = P - B, \tag{22}$$

where

$$P = \frac{1}{N-1}\sum_{r=2}^{N}\frac{1}{r-1}\sum_{i<r} R_{r,i}, \tag{23}$$

$$B = \frac{1}{N-1}\sum_{r=2}^{N}\frac{1}{r-1}\sum_{i<r} R_{i,i}. \tag{24}$$

Both P and B are error levels: the average subsequent error on previously seen tasks and the corresponding weighted initial reference. A high B makes BWT very negative even when P is not especially low. The first task is referenced at every later stage, so for $N = 4$ its coefficient in B is $11/18$, compared with $5/18$ and $1/9$ for the second and third tasks.

For A-first runs, the subsequent-error and reference terms are 1.46 and 13.69 m, yielding BWT $= -12.23$ m despite a final mean error of 1.28 m (Table V). Across algorithms, the correlation of D with the FDE reference term B is between -0.69 and -0.57, whereas its correlation with the subsequent-error term P is only $+0.02$ to $+0.19$. The stronger association with FDE-BWT thus comes largely from the order-dependent reference, and the marginal correlation is not direct evidence that larger D harms retention.

Final error gives a different comparison (Table VIII). SyReM has the lowest uniform-order final FDE (1.08 m) and MR (9.23%) with bootstrap probability 1.00 for both, and its per-order lead is separable in 17 and 16 of the 24 orders, whereas GSM's smallest uniform-order FDE-BWT (-3.33 m) is within 0.03 m of Vanilla, with probabilities 0.51 and 0.46 (Table IX). That lead comes entirely from the six A-first orders, where the first-stage reference is largest: over the 18 remaining orders SyReM has the smallest score in all four metrics, with bootstrap probabilities of 0.90, 0.89, 0.98, and 1.00. Improvement relative to an algorithm-specific initial

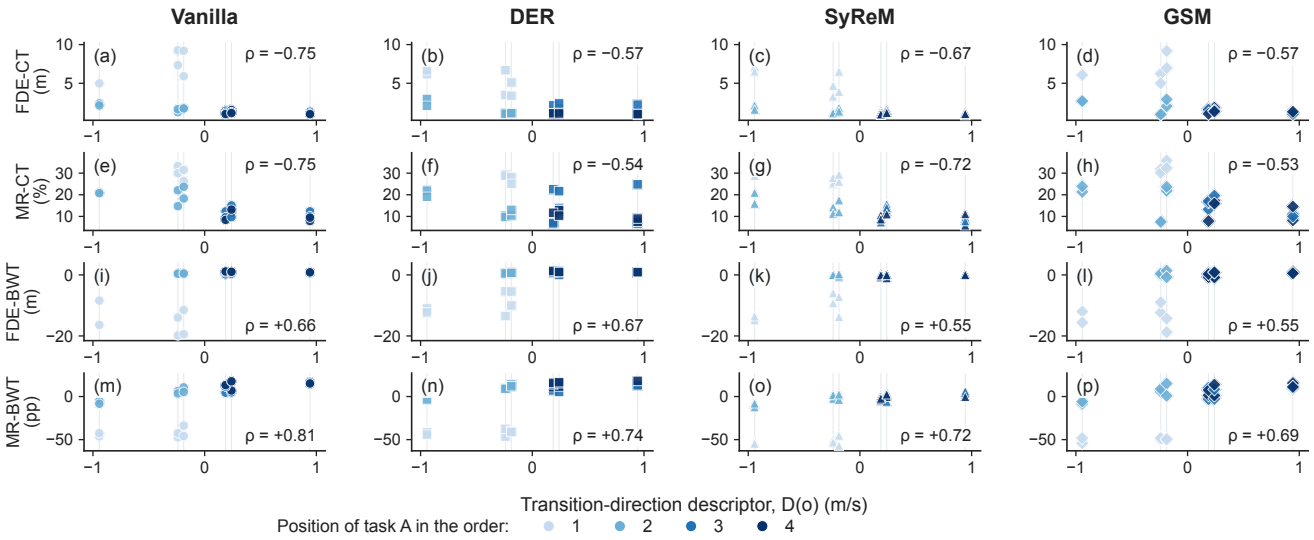


Fig. 5. Transition-direction descriptor versus performance for each algorithm. Points are replicate means for individual orders, coloured by the position of task A; ρ is the Spearman correlation with D. Grey guides mark the six distinct descriptor values.

checkpoint and absolute final prediction quality therefore induce different comparisons, and only the second is decided by these data. CT and BWT contain common initial-error terms with opposite signs, so their opposite associations with D do not by themselves establish an optimisation trade-off.

E. Different Paths from the Same Starting Task

Fixing the first task leaves six orders of the remaining three. Fig. 6 compares the largest- and smallest-L path within each group under the W1 matrix on speed; for every starting task the maximum-distance path has the smaller D. The paired paths differ in measured performance, but not in a direction shared by all metrics (the two B-first paths, for example, have algorithm-averaged FDE-CT values of 2.55 and 1.33 m), which shows sensitivity beyond starting identity without resolving its source. The paths are geometric extremes under L and bound neither the performance range nor the policy scores.

F. Effect of the Order Policy

Table IX compares the uniform and descriptor-weighted scores. Weighting increases the reported CT error for every algorithm: SyReM's FDE-CT changes from 2.33 to 3.01 m and Vanilla's from 2.94 to 3.76 m, and BWT shifts upward, for example from -13.79 to -6.04 percentage points for SyReM's MR-BWT. These changes follow the association used to choose the weighting directions and are not independent confirmation of it. Since 74.2% of the CT weight falls on the 12 orders with A in positions 1–2, the policy mainly reweights the cold-start orders; the effective number of weighted orders is 17.54 of 24.

The complete four-algorithm ranking is identical under the two policies in all four metrics, and the bootstrap probabilities show which leaders are decided by the data: SyReM for MR-CT (0.94) and MR-BWT (1.00), whereas the FDE-CT leader (SyReM, 0.71) and the FDE-BWT leader (GSM, 0.51) are not separable from the runner-up. On this benchmark the weighting therefore shows how aggregate scores move under a stated order emphasis without changing the selection of an algorithm; choosing a policy after seeing which algorithm it favours would introduce a selection effect.

TABLE IX
POLICY SCORES (UNITS AS IN TABLES III AND IV) UNDER THE UNIFORM POLICY, THE DESCRIPTOR-WEIGHTED POLICY, AND THE UNIFORM POLICY OVER THE 18 ORDERS THAT DO NOT START WITH A, WITH THE BOOTSTRAP PROBABILITY OF THE SMALLEST SCORE UNDER THAT POLICY IN PARENTHESES. BOLD: SMALLEST SCORE PER ROW.

Metric	Policy	Vanilla	DER	SyReM	GSM
FDE-CT	Uniform	2.94 (0.00)	2.42 (0.29)	**2.33** (0.71)	2.94 (0.00)
	D-weighted	3.76 (0.00)	3.06 (0.43)	**3.01** (0.57)	3.76 (0.00)
	Excl. A-first	1.43 (0.06)	1.48 (0.04)	**1.31** (0.90)	1.62 (0.00)
MR-CT	Uniform	18.10 (0.00)	17.30 (0.05)	**16.04** (0.94)	19.58 (0.00)
	D-weighted	21.87 (0.00)	20.12 (0.21)	**19.34** (0.79)	22.98 (0.00)
	Excl. A-first	13.86 (0.01)	13.32 (0.09)	**11.98** (0.89)	15.19 (0.00)
FDE-BWT	Uniform	−3.30 (0.46)	−1.88 (0.00)	−2.77 (0.03)	**−3.33** (0.51)
	D-weighted	−1.46 (0.36)	−0.43 (0.00)	−1.23 (0.02)	**−1.55** (0.61)
	Excl. A-first	0.57 (0.00)	0.69 (0.00)	**−0.13** (0.98)	0.12 (0.02)
MR-BWT	Uniform	−3.89 (0.00)	−2.10 (0.00)	**−13.79** (1.00)	−8.17 (0.00)
	D-weighted	5.17 (0.00)	6.01 (0.00)	**−6.04** (1.00)	0.74 (0.00)
	Excl. A-first	9.14 (0.00)	11.13 (0.00)	**−1.00** (1.00)	5.77 (0.00)

VII. DISCUSSION

A. Interpreting the Influence of Task Order

The dominant influence of task order on this benchmark is the learning quality of the first task. Starting-task identity explains 90% to 99% of the order term in every metric, so the across-order component of (6) is essentially a starting-task component, and its source is the cold start of the smallest task (Table V). The first stage involves no cross-task retention mechanism, although the memory-based algorithms do not all behave like Vanilla in it. Once the first stage is removed, the residual order effect is of the same magnitude as training variability, but it is still organised by where task A is placed:

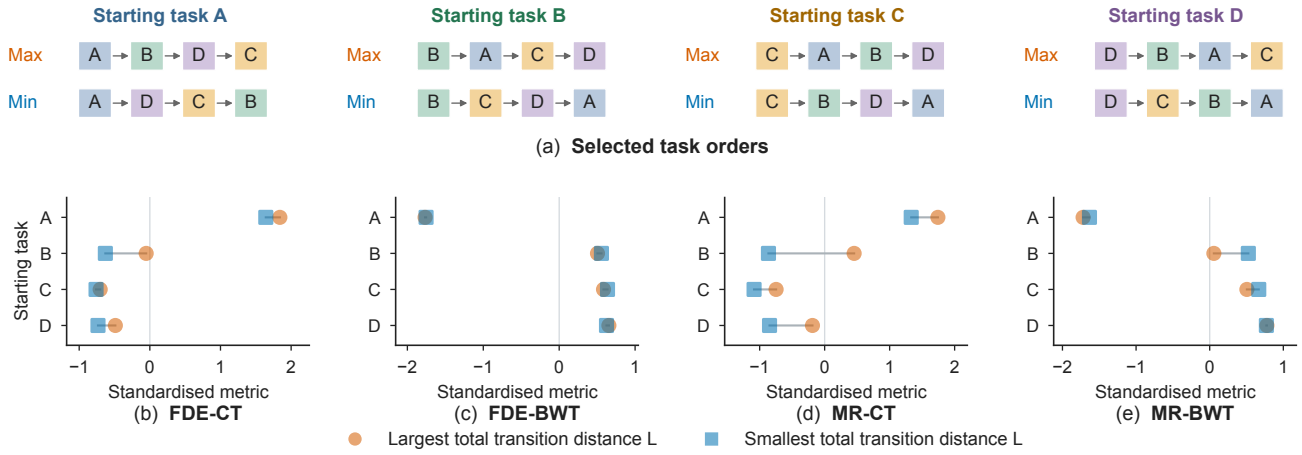


Fig. 6. Paths with the largest and smallest total transition distance L for each starting task. Order strips identify the paths; paired points compare algorithm-averaged scores, standardised separately for each metric across task orders.

the smallest task remains harder to learn at later positions, and the position of A accounts for half to two-thirds of the residual order term.

The transition-direction descriptor inherits this structure. Because the pair containing task A is the more distant pair in every partition of the task set, D is negative exactly when A occupies one of the first two positions, and the position of A correlates more strongly with every primary metric than D does. On this task set, D is best read as an index of when the cold start occurs; what it adds beyond that position is small for the current-task metrics and moderate only for MR-BWT.

The metric decomposition explains why the same first stage moves CT and BWT in opposite directions: the cold-start error enters CT with weight $1/N$, whereas in the reference term of $\mathrm{BWT} = P - B$ the first task carries coefficient $11/18$ for $N = 4$. For A-first orders the reference term is an order of magnitude larger than the subsequent error, so the strongly negative BWT records recovery from a poor start rather than retention, and the order dependence of BWT is largely a reference-term effect.

B. Implications for Algorithm Comparison and Benchmark Design

Four reporting items follow for continual learning benchmarks.

1) State the order distribution behind every aggregate score, the policy q in (4).
2) Report order-resolved and starting-task-resolved outcomes, since changing the order can move a score by far more than changing the algorithm (Tables III and V).
3) Report the warm-start error $\mathrm{CT}_{2:N}$ and the final error alongside CT and BWT.
4) Fix any descriptor and weighting rule before outcomes are examined. The descriptor-weighted policy, retained here as a worked example, mainly re-weights the cold-start orders and leaves the full ranking of the four algorithms unchanged (Table IX).

These data decide some algorithm comparisons and not others. SyReM has the smallest final error and the smallest MR-BWT with bootstrap probability 1.00, leads MR-CT with probability 0.94, and leads all four metrics once the A-first orders are set aside, whereas over all orders the FDE-CT and FDE-BWT leaders are not separable from the runner-up and the per-order leader is separable in only 2 to 11 of the 24 orders for CT and BWT, against 16 or 17 of 24 for final error. The interaction term exceeds its within-cell noise expectation for FDE-CT and FDE-BWT, which indicates that algorithms differ in their sensitivity to order, with Vanilla showing the strongest and GSM the weakest association with D in Table VI; whether memory-based rehearsal dampens order effects in general requires more replicates and more task sets.

C. Extension to Other Task Sets

The magnitudes above describe one benchmark with one dataset, four tasks, one backbone, and three replicates per cell; the protocol does not depend on them. For larger N, sampling from a declared order distribution replaces enumeration, with repeated samples separating coverage uncertainty from training variability; samples should balance starting tasks, which carried most of the order term here, and the warm-start decomposition should accompany CT. Descriptors and weights should be fixed before outcomes are seen, since the primary descriptor came from an exploratory scan of 300 candidates. Task representations beyond vehicle speed, such as distances on the inputs or on backbone features, would allow the protocol to be applied where no application-specific variable is available, and validation across independently constructed task sets would show which findings transfer beyond motion forecasting.

VIII. Conclusion

This study formulates continual learning evaluation as an expectation over a declared distribution of task orders and applies it to motion forecasting across all 24 orders of four tasks, four algorithms, and three replicates. Task order accounts for 65% to 91% of the variation in current-task error

and backward transfer, but almost all of that effect is the identity of the starting task, which acts through the cold start of the smallest task; beyond the first stage, the order effect is comparable to training variability and most reversals of replicate-mean rankings lie within it. The first task dominates the reference term of backward transfer, so relative improvement and absolute final error rank the algorithms differently, and the transition-direction descriptor mainly indexes where the smallest task is placed, leaving the ranking unchanged under descriptor weighting. Benchmarks should therefore state the order distribution behind any aggregate, report order-resolved and starting-task-resolved outcomes with warm-start and final error, and fix any weighting rule before outcomes are examined; validation on independently constructed task sets is required before generalising these findings.

APPENDIX A
ANALYSIS DETAILS AND REPRODUCIBILITY

A. Feature and Distance Reconstruction

Target-agent speed and acceleration channels reconstructed from the trajectory arrays match the cached arrays element by element across all twelve subsets; the release includes feature counts, source hashes, and these verification results. Equation (8) gives the W1 distance. For KL and JS, let c_{ib} count speed observations in bin b of $H = 100$ shared equal-width bins spanning the pooled range; with $\epsilon = 10^{-10}$, the smoothed probabilities are

$$p_{ib} = \frac{c_{ib}/\sum_{h=1}^{H} c_{ih} + \epsilon}{1 + H\epsilon}, \tag{25}$$

and, writing $m_{ijb} = (p_{ib} + p_{jb})/2$,

$$d_{ij}^{\mathrm{KL}} = \sum_{b=1}^{H} p_{ib} \log \frac{p_{ib}}{p_{jb}}, \tag{26}$$

$$d_{ij}^{\mathrm{JS}} = \frac{1}{2} \sum_{b=1}^{H} \left(p_{ib} \log \frac{p_{ib}}{m_{ijb}} + p_{jb} \log \frac{p_{jb}}{m_{ijb}} \right), \tag{27}$$

with natural logarithms; JS denotes the divergence rather than its square root. For MMD, a fixed sample of $n = 2{,}000$ speed observations per task is standardised with the pooled mean and population standard deviation of all retained speed observations, $z_{iu} = (v_{iu} - \mu_v)/\sigma_v$, and compared with the kernel $k(z, z') = \exp[-(z - z')^2]$; the same samples are reused across pairs, and $d_{ij}^{\mathrm{MMD}} = \sqrt{\max(0, Q_{ij})}$, where Q_{ij} is the biased squared discrepancy $n^{-2} \sum_{u,v} [k(z_{iu}, z_{iv}) + k(z_{ju}, z_{jv}) - 2k(z_{iu}, z_{jv})]$ including diagonal terms and the clipping removes roundoff below zero. No kernel two-sample test is performed. MMD samples are drawn without replacement using NumPy PCG64 with seed 0, proceeding through tasks A–D in order. Exact bin edges, standardisation constants, and sampled indices accompany the data. Historical matrices are retained for sensitivity analysis; the original speed-W1 matrix symmetrised as $(M + M^{\mathsf{T}})/2$ induces the same descriptor ranking as the reconstruction. Display scaling is excluded from all calculations.

B. Exploratory Descriptor Scan

Before the five candidates of Section IV-A were fixed, an exploratory scan compared 20 distance representations (W1, KL, JS, and MMD on speed, acceleration, relative distance, and relative speed, plus four multivariate distances) combined with 15 aggregation operators over the three transitions (the individual transitions, their total, mean, minimum, maximum, range, standard deviation, slope, and six further shape summaries) against the same 24-order outcomes (300 candidates; the complete scan table is included with the released inputs). The W1-on-speed slope was retained for its interpretable scale and consistent sign pattern; because this selection used the evaluation outcomes, Table VI is a sensitivity analysis rather than independent validation.

C. Decomposition, Adjustment, and Bootstrap

For a fixed metric, let Z_{oas} be a run outcome with $K = |\mathcal{O}|$ orders, $J = |\mathcal{A}|$ algorithms, and S replicates per combination. The balanced two-way decomposition is $SS_T = SS_O + SS_A + SS_{OA} + SS_W$, with $SS_O = JS \sum_o (\bar{Z}_{o\cdot\cdot} - \bar{Z})^2$, $SS_A = KS \sum_a (\bar{Z}_{\cdot a\cdot} - \bar{Z})^2$, $SS_{OA} = S \sum_{o,a} (\bar{Z}_{oa\cdot} - \bar{Z}_{o\cdot\cdot} - \bar{Z}_{\cdot a\cdot} + \bar{Z})^2$, and $SS_W = \sum_{o,a,s} (Z_{oas} - \bar{Z}_{oa\cdot})^2$, where dots denote averaging and $\bar{Z}$ is the grand mean. Writing $\bar{Z}_{g(o)}$ for the mean over orders with the same starting task as o, the order term splits as $SS_O = JS \sum_o (\bar{Z}_{g(o)} - \bar{Z})^2 + JS \sum_o (\bar{Z}_{o\cdot\cdot} - \bar{Z}_{g(o)})^2$. The within-combination term is invariant to replicate relabelling within each combination, and the fractions are descriptive rather than causal variance components; they are not adjusted for degrees of freedom, so under pure within-cell noise a term with ν degrees of freedom is expected to take ν/ν_W of the within-cell share, with $\nu_W = KJ(S - 1)$ and $\nu = (K - 1)(J - 1)$ for the interaction.

Conditional rank association for an algorithm–metric series assigns average ranks to D and to the order mean, subtracts the respective mean rank within each group, and correlates the two residual vectors; the groups are the four starting tasks or the four positions of task A. The bootstrap resamples the three replicates of every order–algorithm cell with replacement (5,000 draws, NumPy PCG64 seed 0), recomputes replicate means, uniform and weighted scores, and per-order minima, and reports the fraction of draws in which each algorithm attains the smallest score together with 2.5th and 97.5th percentiles of its per-order win count. All weighted means are convex combinations of order means, and the effective number of weighted orders, $(\sum_o w_o)^2 / \sum_o w_o^2$, describes weight concentration only.

D. Training Protocol and Replicates

The released package reproduces the post-training analysis from the saved stage–task results; model training code, configurations, and numerical random seeds are maintained separately by the authors. Replicate labels are local to each order–algorithm cell: in the saved first-stage results, one replicate label gives identical first-stage errors for Vanilla and GSM within each starting task, whereas the other two labels differ across orders with the same starting task, so the

within-cell term mixes replicates with and without a shared initial state. Under the same label, DER and SyReM give first-stage errors that differ from Vanilla's, so their first-stage computation is algorithm dependent; whether output matching and the memory loss act while the memory fills during the first task is determined by the training configuration, not by the released analysis. In the released files, orders are numbered from 0 to 23 in the lexicographic sequence order of Tables III and IV.

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